

Trailing Stop, Break-Even and Position-Sizing Math for Leveraged Crypto/Futures Bots

Introduction

Well-designed trading bots need clear logic for sizing positions, calculating profit and loss (P&L), managing risk and exiting trades. Two key exit tools are **break-even stops** (moving the stop-loss to the entry price once a trade has a small profit) and **trailing stops** (dynamically moving the stop-loss as the market moves in favour of the trade). These mechanisms are particularly important when using leverage, because the percentage move in price is amplified into a much larger percentage change in account equity. The formulas and examples below consolidate the mathematics used in the user-provided trading bot (version 3.4) together with real-world guidance from articles on trading platforms.

Position-Sizing Formulas with Leverage

Trading bots often risk a small percentage of the account on each trade. When trading futures contracts on exchanges such as KuCoin, the trader supplies margin and selects a leverage multiple. The following formulas are used in version 3.4 of the user's bot:

Formula	Description
$\text{marginUsed} = \text{accountBalance} \times (\text{positionPercent} / 100)$	The amount of capital allocated to a single trade. For example, risking 0.5 % of a US\$10 000 account uses US\$50 as margin.
$\text{positionValueUSD} = \text{marginUsed} \times \text{leverage}$	Total exposure. At 10× leverage, US\$50 margin controls US\$500 of notional value.
$\text{size} = \text{floor}(\text{positionValueUSD} / (\text{entryPrice} \times \text{multiplier}))$	Quantity to order. Futures contracts usually have a contract multiplier of 1.
$\text{actualPositionValueUSD} = \text{size} \times \text{entryPrice} \times \text{multiplier}$	Exposure after rounding lot size.
$\text{actualMarginUsed} = \text{actualPositionValueUSD} / \text{leverage}$	Final margin used after rounding lots.

These formulas align with general position-sizing advice. Flipster's guide notes that position size should be **Account Equity × Risk % ÷ |Entry – Stop|**. Leveraged trading compresses the required margin while leaving the risk exposure unchanged, so careful sizing and stop placement are essential.

P&L and Leveraged Percentage

For every tick of price movement, leveraged traders experience a much larger percentage change in their account because the notional exposure is larger than the margin. The bot uses the following formulas:

1. **Price difference:** for long positions, $\text{priceDiff} = \text{currentPrice} - \text{entryPrice}$; for shorts, $\text{priceDiff} = \text{entryPrice} - \text{currentPrice}$.
2. **Unrealised P&L:** $\text{unrealizedPnl} = \text{priceDiff} \times \text{size} \times \text{multiplier}$ (in USDT).
3. **Leveraged P&L %:** $\text{leveragedPnlPercent} = (\text{unrealizedPnl} / \text{marginUsed}) \times 100$.

Using leveraged P&L rather than simple price percentages corrects a common mistake. In a 10× leveraged position, a **0.2 %** price rise translates to a **2 %** profit on the margin. The difference is illustrated below:

Price movement (BTC/USDT)	Leveraged P&L (10×)
+0.1 % price	+1 % margin P&L
+0.5 % price	+5 % margin P&L
+2.0 % price	+20 % margin P&L

The bot therefore computes the leveraged P&L% on every tick and uses it for break-even and trailing logic. This approach is consistent with leveraged trading guides such as CoinGecko's explanation that leverage equals **1 / margin ratio** and that a 10× leveraged long at \$1 000 has a liquidation price at \$900 (a 10 % price move) ¹.

Break-Even Stop Logic

Moving the stop-loss to the entry price once the trade shows a small profit removes downside risk. The bot's break-even logic is:

1. **Trigger:** if $\text{leveragedPnlPercent} > 0.001 \%$ (a tiny profit), then set $\text{currentSL} = \text{entryPrice}$ and record that break-even has been triggered.
2. **Reset trailing baseline:** set $\text{lastTrailingLevel} = \text{leveragedPnlPercent}$ so that subsequent trailing stops are measured from this level.

The break-even threshold uses leveraged P&L rather than price percentage. For example, with 10× leverage the margin is US\$50 and the position size is 0.01 BTC at US\$50 000. Break-even triggers when the P&L exceeds US\$0.0005, which corresponds to a **US\$0.05** price move. Older versions that used a price-based 0.001 % threshold would have moved the stop after a US\$0.50 movement, far too early. The WunderTrading article gives a similar example where a BNB/USDT long at US\$1 000 initially has a stop at US\$990 and moves to break-even after a 3 % price rise (activation) and a 2 % execution price ². After moving the stop to break-even, any retracement does not cause a loss.

Trailing Stop Logic

A **trailing stop** follows the market at a fixed distance. If the price continues to move in the trader's favour, the stop moves; if the price retraces by the distance, the position closes. GoodCrypto's explanation notes that a 10 % trailing stop on BTC bought at US\$10 000 will lock in profits as the price rises. If the price reaches US\$11 111, the stop moves from US\$9 000 to US\$10 000; if the price then falls 10 % to US\$10 000 the position exits, locking in the gain ³.

Bot Implementation (V3.4)

The bot defines two configuration parameters:

- **TRAILING_STEP_PERCENT (0.15 %)** – the minimum increment of *leveraged P&L %* that must be achieved before the stop is trailed again.
- **TRAILING_MOVE_PERCENT (0.05 %)** – the amount (in price %) by which the stop-loss moves for each step.

On every price update, if break-even has been triggered, the bot executes the following logic:

```
profitPercent = leveragedPnlPercent
stepsSinceLastTrail = floor((profitPercent - lastTrailingLevel) /
TRAILING_STEP_PERCENT)
if stepsSinceLastTrail > 0:
    slMovePercent = stepsSinceLastTrail × TRAILING_MOVE_PERCENT
    if side == long:
        newSL = currentSL × (1 + slMovePercent/100)
    else:
        newSL = currentSL × (1 - slMovePercent/100)
    // move stop only in favourable direction
    if (long and newSL > currentSL) or (short and newSL < currentSL):
        currentSL = newSL
        lastTrailingLevel = profitPercent
        updateStopLossOrder(newSL)
```

This algorithm uses **leveraged P&L %** to determine how many “steps” of profit have been gained since the last trail. Each step moves the stop by a small percentage of price, ensuring that profits are locked in progressively but not so aggressively that normal volatility stops the trade. The trailing stop never moves backwards; only increases (for long positions) or decreases (for shorts).

Example – 10× Long Position

- **Entry:** US\$50 000 per BTC; size = 0.01 BTC; margin = US\$50.
- After break-even triggers, the stop is at US\$50 000.
- If price rises to US\$50 075 (a +0.15 % price move), the P&L is US\$0.75 and leveraged P&L % = 1.5 %.
Steps since last trail = $\text{floor}(1.5 / 0.15) = 10$. The stop moves by $10 \times 0.05 \% = 0.5 \%$ of

price: new stop = $50\,000 \times (1 + 0.005) = \text{US\$50 250}$. The trade now locks in a 0.5 % price profit (5 % on margin).

- If price eventually reaches US\$51 000 (+2 % price), leveraged P&L % = 20 %. Steps since last trail = 133, so the stop moves by 6.65 % of price to **US\$53 325**. The trader will capture at least a 6.65 % price gain if the market reverses.

Example – Long Position with Price-Distance Trailing (Pionex)

Pionex's tutorial uses a 1 BTC long at 28 000 USDT with a trailing stop set when price reaches 30 000 USDT and a retracement distance of 1 000 USDT. The minimum stop price is $30\,000 - 1\,000 = 29\,000$. As price rises to 31 000 and 32 000 the trailing stop moves to maintain the 1 000 distance, locking in profit. When price finally retraces by 1 000 USDT, the position is closed at market price, capturing the gains. The example shows how trailing stops move with the highest price and remain stationary during pull-backs until the retracement threshold is met.

Example – Short Position with Percentage Trailing (Pionex)

In another example, a 1 BTC short at 34 500 USDT uses a 10 % retracement rate when price reaches 30 000 USDT. The stop starts at $30\,000 \times (1 + 10\%) = 33\,000$. As price falls, the trailing stop moves down to chase the trend. Once price rallies by 10 % from its lowest point, the stop is hit and the position closes, locking in profit. This illustrates that trailing stops can be defined either by fixed price distance or by percentage.

Risk Management and Leverage Effects

Leverage magnifies both gains and losses. A 10× leveraged position means that a **0.5 %** adverse price movement results in a **5 %** loss on the trader's margin. The bot's default stop-loss percentage (0.5 % price) therefore caps the leveraged loss at around 5 % of the margin. For a US\$50 margin on a US\$10 000 account, the worst-case loss is **US\$2.50**, leaving the account at US\$9 997.50. On the other hand, the take-profit target of 2 % price corresponds to a 20 % gain on the margin (US\$10). This 4:1 risk-reward ratio is a prudent baseline. CoinGecko explains that the liquidation price of a 10× long is roughly 10 % below entry ¹, which underscores the need for conservative stop-losses.

Placing Orders on KuCoin via API

The bot interacts with KuCoin's perpetual futures API using signed REST requests. Key functions in `server.js` wrap the API endpoints:

Function	Purpose	Important parameters
<code>placeOrder(params)</code>	Places a standard order (limit or market).	<code>clientOid</code> (unique ID), <code>side</code> ('buy' for long entries, 'sell' for short entries), <code>symbol</code> (e.g., XBTUSD), <code>type</code> (<code>limit</code> or <code>market</code>), <code>price</code> (for limit orders), <code>size</code> (lot quantity), <code>leverage</code> (e.g., 10), <code>timeInForce</code> (<code>GTC</code> for good-till-cancelled).
<code>placeStopOrder(params)</code>	Places a conditional order used for stops (stop-loss or take-profit).	<code>stop</code> (<code>down</code> for long stops, <code>up</code> for short stops), <code>stopPrice</code> (trigger price), <code>stopPriceType</code> (<code>TP</code>), <code>reduceOnly</code> (true to ensure the order only reduces a position).
<code>cancelOrder(orderId)</code> / <code>cancelStopOrder(orderId)</code>	Cancels existing orders.	Used when updating or closing positions.
<code>cancelAllStopOrders(symbol)</code>	Cancels all open stop orders for a symbol.	

Entry, Stop-Loss and Take-Profit

When a new trade is opened, the bot constructs three orders:

1. **Entry limit order** (via `placeOrder`): sets the entry price and size with the desired leverage. The `clientOid` is prefixed with `entry_`, and `timeInForce` is set to `'GTC'`.
2. **Stop-loss order** (via `placeStopOrder`): a market order conditioned on price hitting the stop level. The bot sets `stop` to `'down'` for long positions (stop triggers below price) or `'up'` for shorts, `stopPrice` to the calculated stop level, `stopPriceType` to `'TP'` (KuCoin's type for stop-loss), `type` to `'market'` and `reduceOnly` to `true` so the order does not open a new position. This order is replaced whenever the stop-loss is moved during break-even or trailing adjustments.
3. **Take-profit order** (via `placeOrder`): a limit order at the take-profit price. It also uses `reduceOnly` to avoid increasing exposure.

Updating the Stop-Loss

Whenever the break-even or trailing logic updates `currentSL`, the bot cancels the previous stop-loss order with `cancelStopOrder`, then submits a new stop order with the updated `stopPrice`. The new stop price is logged and broadcast to the dashboard. This ensures that KuCoin has a single active stop at any time.

Practical Examples Combining the Formulas

Example 1 – Long Position Break-Even and Trailing

Suppose the bot opens a long position on BTC/USDT:

- **Account balance:** US\$10 000; **risk:** 0.5 %; **leverage:** 10×
- **Entry:** 1 March 2025 at **US\$50 000** per BTC.
- **Position size:** $\text{margin} = 10\,000 \times 0.5\% = \text{US\$}50$; $\text{position value} = \text{US\$}50 \times 10 = \text{US\$}500$;
 $\text{size} = \text{floor}(500 / 50\,000) = 0.01 \text{ BTC}$.
- **Initial stop-loss:** 0.5 % below entry $\rightarrow 50\,000 \times (1 - 0.005) = \text{US\$}49\,750$. **Take-profit:** 2 % above entry $\rightarrow \text{US\$}51\,000$.

After the trade is filled, the price drifts up to **US\$50 025**. The margin P&L is $(50\,025 - 50\,000) \times 0.01 = \text{US\$}0.25$, representing $(0.25 / 50) \times 100 = 0.5\%$ leveraged profit. Because $0.5\% > 0.001\%$ (the break-even trigger), the bot moves the stop-loss to **US\$50 000** (break-even) and records the current leveraged P&L % as the trailing baseline.

As price continues to rise, each additional 0.15 % of leveraged profit causes the stop to move up by 0.05 % of price. For example, when the leveraged P&L reaches 1.5 % (corresponding to a 0.15 % price rise), the stop moves up by 0.5 % price. If the price then reverses, the stop is hit at a profitable level, protecting the trader from a draw-down.

Example 2 – Short Position with Percentage Trailing

A trader shorts BTC/USDT at **US\$34 500** with a 10 % retracement rate. When the price reaches **US\$30 000**, the trailing stop is set with a retracement distance of 10 %. The stop starts at $30\,000 \times (1 + 10\%) = \text{US\$}33\,000$. As the price falls, the stop adjusts downward. When the price eventually bounces 10 % from the lowest point, the trailing stop order is triggered, closing the position at market price. This scenario shows how percentage-based trailing stops can lock in profits on short trades.

Example 3 – KuCoin Trailing Order Parameters

KuCoin's interface allows traders to set a **trailing stop loss** with three parameters: an **activation price**, a **trailing delta** and a **price target**. Once both the activation threshold and the retracement threshold are met, KuCoin automatically places a limit order at the target price ⁴. Although the bot implements trailing stops manually via stop orders, understanding KuCoin's native trailing order helps with designing user interfaces.

Best-Practices Checklist

The following checklist summarises the critical checks from version 3.4 of the bot:

- **Position sizing:** verify that $\text{marginUsed} = \text{accountBalance} \times \text{positionPercent}$;
 $\text{positionValue} = \text{marginUsed} \times \text{leverage}$; and $\text{size} = \text{positionValue} \div \text{entryPrice}$.
Round size to the nearest allowed lot.

- **P&L calculation:** always use the leveraged P&L % ($\text{unrealizedPnl} / \text{marginUsed} \times 100$) rather than raw price percentages.
- **Break-even:** trigger when leveraged P&L % > 0.001 %; move the stop to `entryPrice` exactly.
- **Trailing:** use leveraged P&L % to determine steps (0.15 % increments); move the stop by 0.05 % of price per step; never move the stop backwards.
- **Risk management:** choose stop-loss and take-profit levels such that the risk on margin is acceptable (e.g., 5 % loss vs. 20 % gain at 10× leverage); adjust position percent and leverage according to volatility and account size.
- **API parameters:** specify `clientOid` for uniqueness; use `reduceOnly` for stops and take-profit; set `stop` to `'down'` for long stops and `'up'` for shorts; always cancel old stop orders before placing new ones.

Conclusion

Trailing stops and break-even adjustments allow a leveraged trading bot to lock in profits while letting winners run. Correctly calculating position size, leveraged P&L and risk ensures that the bot behaves predictably across instruments and leverage levels. The formulas and examples in this reference are drawn from the user's bot and supported by external sources such as Flipster's position-sizing guide, GoodCrypto's explanation of trailing stops ³, WunderTrading's break-even examples ² and CoinGecko's leverage overview ¹. They provide a solid foundation for designing indicator-driven entry strategies and for generating signals that the bot can act on via the KuCoin API.

¹ Leverage Crypto Trading: How Does It Work? | CoinGecko

<https://www.coingecko.com/learn/leverage-crypto-trading-how-does-it-work>

² How Move Stop Loss to Breakeven Works | Help Center

<https://help.wundertrading.com/en/articles/9250110-how-move-stop-loss-to-breakeven-works>

³ Trailing Stop Loss and Trailing Take Profit Orders Explained

<https://goodcrypto.app/trailing-stop-loss-and-trailing-take-profit-orders-explained/>

⁴ KuCoin Review 2025: Exchange With +30 Million Users!

<https://tradingfinder.com/exchanges/kucoin/>